

Optic disc vessel density changes after shunt surgery in idiopathic intracranial hypertension

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ARTICLE INFO

Keywords:

Idiopathic intracranial hypertension
Optic disc vessel density
Shunt surgery
OCTA

ABSTRACT

Aim: To detect the change in optic disc vessel density in patients with idiopathic intracranial hypertension (IIH) after shunt surgery.

Patients and method: Twenty-one patients with IIH underwent shunt surgery. Patients were subjected to full ophthalmic and neurological examinations. Optic disc vessel density was measured using optical coherence tomography angiography (OCT-A) before and two months after surgery.

Results: The age of patients ranged from 27 to 48 years old, with a mean of (34.85 ± 6.87) . Visual acuity ranged between 0.2 and 0.9 by log Mar, with a mean of (0.64 ± 0.20) . The mean deviation (MD) of the perimeter ranged from -22.9 to 0 dB, with a mean of (-9.77 ± 7.23) dB. With respect to the grade of papilledema, 15 eyes (35.7%) were of grade II, 21 eyes (50.0%) were of grade III, and 6 eyes (14.3%) were grade IV.

Intracranial pressure measured by lumbar puncture (LP) prior to surgery ranged from 30 to 40 cm, with a mean of 35.43 cm. Except for right transverse sinus hypoplasia in 6 patients (28.6%), MRV was normal. In 15 patients (71.4%), lumboperitoneal shunt was performed, while ventriculoperitoneal shunt was performed in 6 patients (28.6%).

After surgery, the optical disc vessel density was decreased in all patients. This decrease was highly significant for the whole image, the inner disc, the peripapillary, superior, and inferior hemispheres. Moreover, this reduction affected both optic disc microvasculature (referred to as capillary vessel density) and macro vasculature (also known as all vessel density).

There were positive correlations between the CSF opening pressures and the preoperative optic disc vessel density of the whole image and inside disc with a P-value (0.002, 0.009), respectively. In addition, there was a positive correlation between the opening CSF pressures and the reduction in whole image vessel density after surgery with a P-value of 0.000.

Conclusion: Optic disc vessel density decreased after shunt surgery in patients with IIH. Consequently, optic disc vessel density can be used as a biomarker for the reduction of CSF opening pressure.

1. Introduction

Idiopathic intracranial hypertension (IIH) is a condition of benign increase in intracranial tension characterized by a severe progressive headache that may be associated with vomiting and vision blurring. Increased intracranial tension is demonstrated by an LP of >25 mmHg [1]. The presentation involves papilledema with no intracranial space-occupying lesion, normal brain MRI, and signs of IIH. Optic nerve head protrusion, increased perioptic cerebrospinal fluid (CSF), optic

nerve tortuosity, posterior sclera flattening, empty sella, bilateral transverse venous sinus stenosis, cerebella tonsillar descent, cephaloceles, and enlarged Meckel caves are MRI indicators of intracranial hypertension [2–4]. The incidence of IIH in the Middle East is 2.02–2.2/100,000 [2]. If medical treatment is unsuccessful, this condition may be managed surgically. Several surgical procedures can be performed, including ventriculoperitoneal and lumboperitoneal shunts. In addition, some patients can undergo subtemporal decompression and optic nerve sheath fenestration [5].

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<https://doi.org/10.1016/j.pdpdt.2023.103625>

Received 8 March 2023; Received in revised form 17 May 2023; Accepted 19 May 2023

Available online 27 May 2023

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Table 1
compression between the pre- and postoperative optic disc vessel density.

Optic disc vessel density%		Preoperative	Postoperative	Difference Mean $\pm$ SD	Test value	P-value	Sig.
Whole image(cap)	Mean $\pm$ SD	54.26 $\pm$ 2.49	48.02 $\pm$ 2.45	-6.24 $\pm$ 1.95	20.711●	0.000	HS
	Range	49.9 – 58.5	43.4 – 51.4				
Whole image (all)	Mean $\pm$ SD	59.75 $\pm$ 1.52	54.56 $\pm$ 2.25	-5.19 $\pm$ 1.51	22.315●	0.000	HS
	Range	56.2 – 62.3	50.4 – 57.2				
Inside disc (cap)	Mean $\pm$ SD	56.17 $\pm$ 4.73	47.24 $\pm$ 3.54	-8.94 $\pm$ 5.98	9.676●	<0.001	HS
	Range	49.7 – 68.4	41.3 – 53.2				
Inside disc (all)	Mean $\pm$ SD	64.98 $\pm$ 4.68	57.54 $\pm$ 3.38	-7.37 $\pm$ 4.92	9.364●	<0.001	HS
	Range	59.1 – 76.7	51.8 – 62.5				
Peripapillary (cap)	Mean $\pm$ SD	56.19 $\pm$ 2.99	50.44 $\pm$ 3.11	-5.74 $\pm$ 1.48	25.114●	0.000	HS
	Range	50.2 – 60.1	45.1 – 55				
Peripapillary (all)	Mean $\pm$ SD	61.46 $\pm$ 1.91	55.09 $\pm$ 3.98	-6.37 $\pm$ 2.36	17.468●	0.000	HS
	Range	58.4 – 64.4	44.9 – 60.3				
Superior- hemi (cap)	Mean $\pm$ SD	55.36 $\pm$ 3.73	49.54 $\pm$ 4.07	-5.82 $\pm$ 1.26	29.979●	0.000	HS
	Range	46.5 – 60.6	41.2 – 57.2				
Superior- hemi (all)	Mean $\pm$ SD	61.21 $\pm$ 2.87	55.92 $\pm$ 3.40	-5.29 $\pm$ 1.75	19.616●	0.000	HS
	Range	54.3 – 65.7	48.5 – 60.6				
Inferior-hemi (cap)	Mean $\pm$ SD	56.06 $\pm$ 2.77	49.92 $\pm$ 2.92	-6.14 $\pm$ 1.93	20.662●	<0.001	HS
	Range	50.3 – 59.9	45.3 – 54.2				
Inferior- hemi (all)	Mean $\pm$ SD	61.63 $\pm$ 2.02	55.64 $\pm$ 4.11	-5.99 $\pm$ 2.76	14.064●	0.000	HS
	Range	58.4 – 65.8	44.7 – 60.2				

P-value for controlled Family-wise error rate (FWER) using Bonferoni was 0.05/10 = 0.005.

Using: Paired t-test.

Vision is impaired when intracranial pressure is elevated. It primarily affects the optic nerve. However, it can also affect the retina, leading to optic atrophy and blindness. Increased intracranial pressure around the optic nerve sheath leads to axoplasmic flow stasis, resulting in papilledema and axon swelling. In contrast, secondary vascular changes occur in the form of venous congestion and optociliary shunt, whereas compression of axons leads to optic nerve gliosis [6].

If gliosis occurs, papilledema resolution is difficult to detect clinically. Therefore, we believe that the detection of papilledema resolution depends on the vascular changes that occur in papilledema. This can be accomplished by measuring the vessel density of the optic disc using optical coherence tomography angiography (OCT-A). OCT-A is a non-invasive quick technique. It provides quantitative, reproducible measurements of vessel density. It can measure the vessel density of the optic disc [7].

In this study, we aimed to determine the availability of optic disc vessel density as a quantitative method to monitor intracranial pressure.

2. Patients and methods

The prospective study included 21 patients with IIH with different grades of papilledema. It was done between January and November 2022. All patients had to sign informed consent. The study was performed in accordance with the Declaration of Helsinki. All patients were diagnosed and operated on in a tertiary university hospital.

The patients included in this study had IIH and were scheduled to undergo surgery. Patients who were not scheduled for surgery were not included in the study. Other exclusion criteria included patients with myopia greater than -6 D, any systemic disease affecting the eye, history of ocular trauma or surgery, medications that affect ocular blood vessels, alcohol intake, pregnancy, smoking, and any condition that demonstrates inaccurate OCT-A results. Moreover, preoperatively patients with IIH with no papilledema were excluded from the study.

All patients were subjected to full history taking, fundus examination, neurological examination, MRI brain, and MR venography. LP was performed in the lateral decubitus position to determine the CSF opening pressure, in addition to CSF analysis.

The ophthalmological examination included visual acuity measurement, anterior segment examination, intraocular pressure measurement by air puff tonometer, fundus examination by 90 D lens, visual field analysis using Humphrey Field analyzer; (24-2 SITA; Carl-Zeiss

Meditec), and optic disc vessel density measurements using Spectral-domain OCT-A (Optovue Angiovue System, software ReVue XR version 2017.1.0.151).

- The grade of papilledema was classified by the Modified Frisén Scale as follows:
 - Grade 0 (Normal Optic Disc): no papilledema.
 - Grade 1 (Minimal Degree of oedema): C-shaped halo (mainly nasally) that obscures retinal details sparing the temporal disc margin.
 - Grade 2 (Low Degree of oedema): Circumferential halo, an elevation of the nasal border, no obscuration of a major blood vessel.
 - Grade 3 (Moderate Degree of oedema): Circumferential halo with obscuration of one segment of a major blood vessel and an elevation of all borders.
 - Grade 4 (Marked Degree of oedema): Total obscuration of a major blood vessel segment with an elevation of all the optic discs, including the cup.
 - Grade 5 (Severe Degree of oedema): Obscuration of all vessels on the disc.
- OCT-A measurements:

The optical disc vessel densities of all patients were measured prior to and two months after surgery. It was conducted between 9 and 12 a. m. to prevent diurnal variation. Images were taken using Spectral-domain OCT-A (Optovue Angiovue System, software ReVue XR version 2017.1.0.151). The device employs a split-spectrum amplitude decorrelation algorithm (SSADA) to decrease motion errors and improve the signal-to-noise ratio.

Angio disc can protocol was used (4.5 $\times$ 4.5 mm). An ellipse to the disc margin was fitted automatically. The software calculated the average vessel density within the disc (inside disc vessel density). The radial peripapillary vessel density was calculated using a 750 μ m diameter region surrounding the disc. The vessel density was analyzed automatically in various areas (whole image, inside disc, peripapillary, superior-hemi, and inferior-hemi). OCT-A can measure the microvasculature (referred to as capillary vessel density) and macro vasculature (referred to as all vessel density).

Table 2

The relation of preoperative optic disc vessel density among different grades of papilledema.

Preoperative optic disc vessel density%		Grade of papilledema			Test value	P-value	Sig.
		Grade II No. = 15	Grade III No. = 21	Grade IV No. = 6			
Whole image(cap)	Mean $\pm$ SD	58.65 $\pm$ 1.04	60.36 $\pm$ 3.53	62.56 $\pm$ 1.86	6.908●	0.003	HS
	Range	57.70 – 59.60	54.30 – 63.80	59.70 – 65.70			
Whole image (all)	Mean $\pm$ SD	58.66 $\pm$ 1.57	60.46 $\pm$ 1.24	60.00 $\pm$ 0.33	8.522●	0.001	HS
	Range	56.20 – 60.40	58.70 – 62.30	59.70 – 60.30			
Inside disc (cap)	Mean $\pm$ SD	53.22 $\pm$ 3.30	56.27 $\pm$ 2.97	63.20 $\pm$ 5.70	16.972●	0.000	HS
	Range	49.7 – 57.6	51.2 – 60.3	58 – 68.4			
Inside disc (all)	Mean $\pm$ SD	61.76 $\pm$ 2.78	65.64 $\pm$ 3.27	70.70 $\pm$ 6.57	13.089●	0.000	HS
	Range	59.1 – 65.8	59.4 – 69.5	64.7 – 76.7			
Peripapillary (cap)	Mean $\pm$ SD	53.84 $\pm$ 3.81	55.88 $\pm$ 2.91	58.95 $\pm$ 1.37	5.900●	0.006	HS
	Range	50.1 – 59.7	50.5 – 60.1	57.2 – 60.2			
Peripappillary (all)	Mean $\pm$ SD	59.80 $\pm$ 1.53	61.04 $\pm$ 1.96	62.23 $\pm$ 1.62	5.202●	0.010	S
	Range	58.40 – 61.20	58.50 – 63.60	59.40 – 64.40			
Post Hoc analysis by LSD							
		Grade II Vs grade III	Grade II Vs grade IV	Grade III Vs grade IV			
Whole image(cap)		0.461	0.010	0.002			
Whole image (all)		0.000	0.081	0.209			
Inside disc (cap)		0.015	0.000	0.000			
Inside disc (all)		0.004	0.000	0.005			
Peripapillary (cap)		0.061	0.002	0.040			
Peripappillary (all)		0.725	0.013	0.019			

P-value > 0.05: Non significant; P-value < 0.05: Significant; P-value < 0.01: Highly significant.

Using: One Way ANOVA test and Post Hoc analysis by LSD.

2.1. Follow-up

Patients were followed up in the outpatient Clinic 2 weeks, one month, and two months postoperatively. The patient reported whether or not his symptoms (headache and blurred vision) had subsided. Additionally, clinical follow-up and comprehensive ophthalmological examinations of the optic nerve, visual acuity, fundus examination, and visual fields were performed at two months. OCT-A was done for follow-up of papilledema and optic disc vessel density.

3. Statistical analysis

Data were collected, revised, coded, and entered into the Statistical Package for Social Science (IBM SPSS) version 23. The quantitative data

were presented as mean, standard deviations, and ranges when parametric. The qualitative variables were presented as numbers and percentages.

The comparison between groups regarding qualitative data was made using the Chi-square test, and the quantitative data with parametric distribution was done using the Independent *t*-test.

A paired *t*-test was used to compare two paired groups regarding quantitative data and parametric distribution. One Way ANOVA test was used to compare more than two groups regarding quantitative data and parametric distribution followed by Post Hoc analysis using LSD test. *P*-value for controlled Family-wise error rate (FWER) using Bonferroni was done for Table 1 and FWER was calculated.

Spearman correlation coefficients were used to assess the correlation between two quantitative parameters in the same group.

Table 3

The relation of the difference in optic disc vessel density after surgery amongst different grades of papilledema.

The difference in the optic disc vessel density after surgery		Grade of papilledema			Test value	P-value	Sig.
		Grade II No. = 15	Grade III No. = 21	Grade IV No. = 6			
Whole image(cap)	Mean $\pm$ SD	−5.76 $\pm$ 1.47	−5.06 $\pm$ 1.36	−9.80 $\pm$ 5.26	10.061●	0.000	HS
	Range	−8.20 – −4.30	−6.30 – −2.00	−14.60 – −5.00			
Whole image (all)	Mean $\pm$ SD	−4.46 $\pm$ 1.05	−5.29 $\pm$ 1.34	−6.65 $\pm$ 2.03	5.679●	0.007	HS
	Range	−5.80 – −3.20	−6.70 – −3.40	−8.50 – −4.80			
Inside disc (cap)	Mean $\pm$ SD	−6.46 $\pm$ 3.54	−7.67 $\pm$ 2.73	−19.55 $\pm$ 8.27	23.432●	0.000	HS
	Range	−13.1 – −3.6	−13.9 – −5.6	−27.1 – −12			
Inside disc (all)	Mean $\pm$ SD	−5.54 $\pm$ 1.07	−6.37 $\pm$ 1.95	−15.45 $\pm$ 9.37	17.004●	0.000	HS
	Range	−7.1 – −4.2	−9.4 – −3.3	−24 – −6.9			
Peripapillary (cap)	Mean $\pm$ SD	−6.38 $\pm$ 1.23	−5.10 $\pm$ 1.56	−6.40 $\pm$ 0.77	4.657●	0.015	S
	Range	−8.40 – −5.10	−6.80 – −2.40	−7.10 – −5.70			
Peripappillary (all)	Mean $\pm$ SD	−6.64 $\pm$ 1.24	−5.51 $\pm$ 1.00	−8.70 $\pm$ 5.26	5.311●	0.009	HS
	Range	−8.10 – −4.60	−7.30 – −3.90	−13.50 – −3.90			
Post Hoc analysis by LSD							
		Grade II Vs grade III	Grade II Vs grade IV	Grade III Vs grade IV			
Whole image(cap)		0.153	0.045	0.279			
Whole image (all)		0.103	0.000	0.005			
Inside disc (cap)		0.392	0.000	0.000			
Inside disc (all)		0.509	0.000	0.000			
Peripapillary (cap)		0.131	0.000	0.004			
Peripappillary (all)		0.840	0.023	0.025			

P-value > 0.05: Non significant; P-value < 0.05: Significant; P-value < 0.01: Highly significant.

Using: One Way ANOVA test and Post Hoc analysis by LSD.

Table 4

The correlation CSF opening pressure and preoperative optic disc vessel density and the difference in optic disc vessel density after surgery.

	CSF opening pressure	
	r	P-value
Preoperative optic disc vessel density		
Whole image(cap)	0.464**	0.002
Whole image (all)	-0.137	0.388
Inside disc (cap)	0.396**	0.009
Inside disc (all)	-0.077	0.626
Peripapillary (cap)	0.189	0.230
Peripapillary (all)	0.269	0.085
Difference in optic disc vessel density after surgery		
Whole image(cap)	0.527**	0.000
Whole image (all)	-0.014	0.932
Inside disc (cap)	-0.269	0.085
Inside disc (all)	0.191	0.225
Peripapillary (cap)	0.121	0.443
Peripapillary (all)	0.037	0.818

P-value > 0.05: Non significant; P-value < 0.05: Significant; P-value < 0.01: Highly significant.

Using: Spearman correlation coefficient.

4. Results

The study included 42 eyes of 21 patients. The age of patients ranged from 27- 48 years old, with a mean of (34.85 ± 6.87) . Visual acuity ranged from 0.2 to 0.9 by log Mar, with a mean of (0.64 ± 0.20) . The mean deviation (MD) of the perimeter ranged from -22.9 to 0 dB, with a mean of $(-9.77 \pm 7.23$ dB). With respect to the grade of papilledema, 15 eyes (35.7%) were of grade II, 21 eyes (50.0%) were of grade III, and 6 eyes (14.3%) were of grade IV.

LP measurements of preoperative intracranial pressure were from 30 to 40 cm, with a mean of 35.43 cm. Except for right transverse sinus hypoplasia in 6 patients (28.6%), MRV was normal. In 15 patients (71.4%), a lumboperitoneal shunt was performed, while a ventriculoperitoneal shunt was performed in 15 patients (71.4%) and a ventriculoperitoneal shunt in 6 patients (28.6%).

After surgery, the optic disc vessel density was decreased in all patients. This reduction was highly significant either in the whole image, inside disc, peripapillary, superior hemisphere, and inferior hemisphere. Moreover, this reduction affected both optic disc microvasculature (referred to as capillary vessel density) and macro vasculature (referred to as all vessel density) (Table 1). FWER $[1-(1-0.005)^{10}]$ was 0.04.

Regarding papilledema grades, there were significant differences in preoperative optic nerve vessel density (optic disc vessel density increased as the papilledema grade increased) (Table 2).

Moreover, there were statistically significant differences in optic disc

vessel density reduction after surgery in different grades of papilledema (the reduction in optic disc vessel density increased as the papilledema grade increased) (Table 3).

There were positive correlations between the CSF opening pressures and the preoperative optic disc vessel density of the whole image and inside disc with P-values of (0.002 and 0.009), respectively (Table 4) (Fig. 1).

Additionally, there was a positive correlation between the opening CSF pressures and the reduction in whole image vessel density after surgery, with a P-value of 0.000 (Table 4) (Fig. 2).

No correlation was found between the total postoperative RNFL thickness and the postoperative optic disc vessel density. Moreover, there was no correlation between the total postoperative RNFL thickness and the difference in optic disc vessel density (Table 5).

5. Discussion

IIH leads to papilledema in most cases, which affects vision. In addition, it leads to optic neuropathy due to chronic or acute loss of the peripheral retinal nerve fibre layer and the infarction of the retinal nerve fibre. Papilledema can also lead to retinopathy, including hemorrhages, choroidal folds, subretinal fluid, and macular oedema [8]. Venous congestion occurs in papilledema. It can be explained by pressure on the central retinal vein as it has a subarachnoid course. However, the venous congestion is also demonstrated by the axonal swelling that occurs

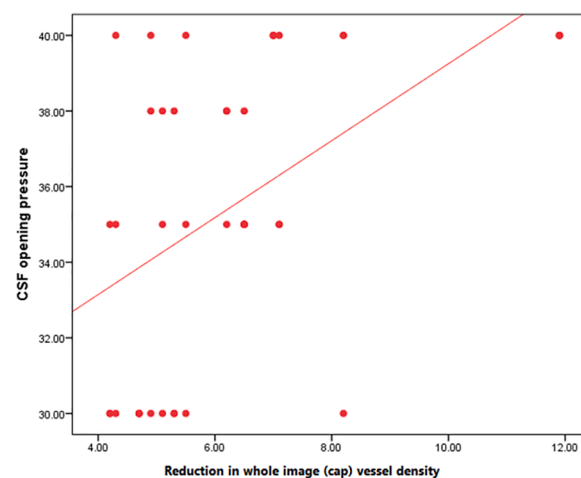


Fig. 2. A positive correlation between CSF opening pressure and the reduction in optic disc whole image (cap) vessel density.

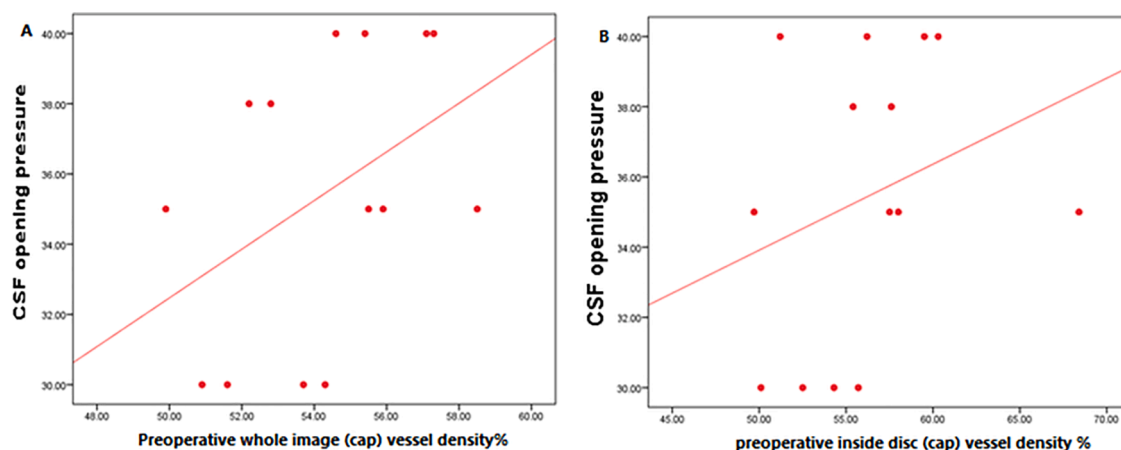


Fig. 1. The correlations between CSF opening pressure and preoperative optic disc vessel density. (A) A positive correlation between CSF opening pressure and whole preoperative image (cap) vessel density. (B) A positive correlation between CSF opening pressure and preoperative inside disc (cap) vessel density.

Table 5

The correlation between the postoperative total RNFL thickness and post-operative optic disc vessel density and the difference in optic disc vessel density after surgery.

	Post-operative total RNFL thickness	P-value
<i>Postoperative optic disc vessel density</i>		
Whole image(cap)	-0.169	0.285
Whole image (all)	-0.129	0.414
Inside disc (cap)	0.051	0.747
Inside disc (all)	0.069	0.664
Peripapillary (cap)	-0.044	0.782
Peripapillary (all)	0.244	0.120
Superior- hemi (cap)	-0.104	0.512
Superior- hemi (all)	-0.073	0.647
Inferior-hemi (cap)	-0.261	0.095
Inferior- hemi (all)	-0.094	0.552
<i>The Difference between pre and post-operative optic disc vessel density</i>		
Whole image(cap)	-0.090	0.571
Whole image (all)	0.085	0.592
Inside disc (cap)	0.040	0.803
Inside disc (all)	0.061	0.701
Peripapillary (cap)	0.088	0.579
Peripapillary (all)	0.180	0.253
Superior- hemi (cap)	-0.108	0.497
Superior- hemi (all)	-0.082	0.606
Inferior-hemi (cap)	-0.021	0.893
Inferior- hemi (all)	-0.255	0.103

P-value > 0.05: Non significant; P-value < 0.05: Significant; P-value < 0.01: Highly significant.

Using: Spearman correlation coefficient.

secondary to axoplasmic flow stasis [9].

We cannot solely depend on headaches as the only indicator of shunt failure. Sinclair et al. [10], who performed an extensive study on IIH, determined that headaches did not improve in 19% of patients. They recommended not depending on headaches as the only sign of shunt failure. Moreover, they [10] discovered that only 44% of patients had their papilledema resolved. Therefore, papilledema cannot be considered an indicator of shunt failure.

The Modified Frisén Scale was utilized for the clinical grading of papilledema. This scale has inter- and intra-observer variability and requires expert evaluation. The presence of gliosis in the optic nerve makes it difficult for even an expert to detect the resolution of papilledema [11]. Furthermore, fundus photography is subjective, and the photo artefact makes interpretation difficult [12]. For all these reasons, we need a qualitative non-invasive method to detect the improvement of intracranial pressure without performing LP. OCT-A is a quantitative and non-invasive method for detecting the optic disc vessel density. It can measure both the optic disc micro and macrovasculature [7].

We found a highly significant reduction in optic disc vessel density in both micro- and macrovasculature. This reduction increased as the grade of papilledema increased. The reduction after surgery increased if the preoperative CSF opening pressure increased. The reduction in optic disc vessel density can be explained by the relief of venous congestion following shunt surgery to reduce CSF pressure. This reduction, most probably, is not due to progressive thinning of RNFL secondary to progressive optic nerve atrophy. This can be emphasized by the absence of a correlation between the total postoperative RNFL thickness and the postoperative optic disc vessel density. Furthermore, the absence of a correlation between the total postoperative RNFL thickness and the difference in optic disc vessel density reinforced this.

Based on the previous results, we can conclude that optic disc vessel density can be used as a quantitative method to detect the success of shunt surgery. However, there are some limitations to using OCT-A, such as in high myopic patients, medications that affect ocular blood vessels,

alcohol intake, pregnancy, and smoking (because these factors affect the optic disc vessel density). However, we can depend on the difference between pre-and postoperative measurements. Optic disc vessel density cannot be measured if dense media opacity hinders light from reaching the retina. In these cases, fundus examination and fundus photography are also challenging, and we must rely on ultrasound imaging of the optic disc. In addition, the lack of patient fixation results in poor image quality and the inability to quantify vessel density. In patients with grade 5 papilledema, it was difficult to obtain high-quality images. This may be another limitation of using OCT-A to measure the vessel density of the optic disc in papilledema.

To our knowledge, this is the first study to detect the change in optic disc vessel density after shunt surgery. The limitation of our study was the small sample size. Therefore, further studies with a larger number of patients should be done. Subgroups with different degrees of papilledema should be investigated thoroughly.

In conclusion, optic disc vessel density can be used as a non-invasive quantitative method to determine the success of shunt surgery, particularly when clinical detection of papilledema is difficult, even for experts. Therefore, the vessel density of the optic disc can be used as a biomarker to reduce CSF opening pressure.

Funding and financial support

None.

CRediT authorship contribution statement

Nermien Salah El-Dien Mohammed El-Haddad: Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – original draft, Writing – review & editing. **Shymaa Adel Ismael:** Visualization, Methodology, Data curation, Writing – original draft. **Adel Abd El-Wahab:** Supervision. **Sawssan Shalaby:** Supervision. **Mona Mohammad Aly Farag:** Supervision. **Nesma Sied Mohammad:** Supervision. **Shrief Shawky:** Supervision.

Declaration of Competing Interest

None.

Acknowledgements

None.

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